Walchand Institute of Technology, Solapur
(An Autonomous Institute)

Affiliated to
Punyashlok Ahilyadevi Holkar Solapur University, Solapur

Choice Based Credit System (CBCS)

Structure and Syllabus
for
M. Tech. Mechanical (Design Engineering)
Scheme 2024

F. Y. M. Tech. Mechanical (Design Engineering)
2026-27

Dr. S. B. Tuljapure
Chairman - Board of Studies

Dean (Academics)

Mechanical Engineering

Vision

- To produce world class globally competent distinguished graduates/ post graduates/ doctoral, Mechanical Engineers on the basis of their capabilities, dedication and work ethic and continuously strive towards societal development.

Mission

- **M1:** To impart quality Mechanical Engineering education in accordance with the needs of the society.
- **M2:** To produce globally competent Mechanical Engineers through research, industry institute interaction.
- **M3:** To help Mechanical Engineering graduates to implement their acquired engineering knowledge for society and community development.

Department of Mechanical Engineering	
Knowledge and Attitude Profile (WK)	
WK1	A systematic, theory-based understanding of the natural sciences applicable to the discipline and awareness of relevant social sciences.
WK2	Conceptually-based mathematics, numerical analysis, data analysis, statistics and formal aspects of computer and information science to support detailed analysis and modelling applicable to the discipline.
WK3	A systematic, theory-based formulation of engineering fundamentals required in the engineering discipline.
WK4	Engineering specialist knowledge that provides theoretical frameworks and bodies of knowledge for the accepted practice areas in the engineering discipline; much is at the forefront of the discipline.
WK5	Knowledge, including efficient resource use, environmental impacts, whole-life cost, re-use of resources, net zero carbon, and similar concepts, that supports engineering design and operations in a practice area.
WK6	Knowledge of engineering practice (technology) in the practice areas in the engineering discipline.
WK7	Knowledge of the role of engineering in society and identified issues in engineering practice in the discipline, such as the professional responsibility of an engineer to public safety and sustainable development.
WK8	Engagement with selected knowledge in the current research literature of the discipline, awareness of the power of critical thinking and creative approaches to evaluate emerging issues.
WK9	Ethics, inclusive behaviour and conduct. Knowledge of professional ethics, responsibilities, and norms of engineering practice. Awareness of the need for diversity by reason of ethnicity, gender, age, physical ability etc. with mutual understanding and respect, and of inclusive attitudes.
Program Outcomes (POs)	
PO 1	An ability to independently carry out research /investigation and development work to solve practical problems
PO 2	An ability to write and present a substantial technical report/document
PO 3	An ability to demonstrate a degree of mastery over the area as per the specialization of the program. The mastery should be at a level higher than the requirements in the bachelor program of Mechanical Engineering.
PO 4	An ability to design mechanical systems by using principles of continuum mechanics, vibrations, reliability, maintainability, rapid prototyping

	&reverse engineering methods and feasibility of manufacturing processes.
PO 5	An ability to apply analytical and computational approach by employing software tools to design and develop a novel product and to optimize the existing ones.

Department of Mechanical Engineering

Legends Used

L	Lecture Hours / week
T	Tutorial Hours / week
P	Practical Hours / week
FA	Formative Assessment
SA	Summative Assessment
ESE	End Semester Examination
ISE	In Semester Evaluation
ICA	Internal Continuous Assessment
POE	Practical and Oral Exam
OE	Oral Exam
MOOC	Massive Open Online Course
HSS	Humanity and Social Science
NPTEL	National Programme on Technology Enhanced Learning
F.Y.	First Year
S.Y.	Second Year
T.Y.	Third Year
B. Tech.	Bachelor of Technology

Department of Mechanical Engineering

Course Code Format

M	D	U/P	2	C	C	1
Program Code		U-Under Graduate, P-Post Graduate	Semester No. / Year 1/2/3/4	Course Type		Course Serial No. 1-9

Program Code	
MD	M. Tech Mechanical-Design Engineering
Course Type	
CC	Core Compulsory Course
EN*	Core Elective N* indicates the serial number of electives offered in the respective category
ON* Open Elective	N* indicates the serial number of electives offered in the respective category
SM	Seminar
DS	Dissertation
IN	Internship

Sample Course Code	
MDP1CC1	Advanced Stress Analysis

Mechanical-Design Engineering

M. Tech. Semester I

Course Code	Name of Course	Engagement Hours			Credits	SA		FA		Total
		L	T	P		Theory	OE/ POE	ISE	ICA	
MDP1CC1	Advanced Stress Analysis	3			3	60		40		100
MDP1CC2	Advanced Vibrations and Acoustics	3			3	60		40		100
MDP1CC3	Industrial Instrumentation	3		2	4	60		40	25	125
CMP1RM	Research Methodology and IPR	3	1		4	60		40	25	125
MDP1CE1N*	Core Elective- I	3		2	4	60		40	25	125
MDP1SM1	Seminar-I			4	2				50	50
	Grand Total	**15**	**1**	**8**	**20**	**300**		**200**	**125**	**625**

Note:

- N*indicatestheserialnumberofelectivesofferedintherespectivecategory

Course Code	***Name of the Course***	***Engagement Hours***			***Credits***	***SA***	***FA***		***Total***
		L	**T**	***P***		***ESE***	***ISE***	***ICA***	
CMP1AC	Yoga for Stress Management	2					50		50

Core Elective-I

Course Code	Course Title
MDP1CE11	Reliability Engineering
MDP1CE12	Mechanical System Design
MDP1CE13	Computer Aided Design

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Mechanical-Design Engineering

M. Tech. Semester II

Course Code	Name of Course	Engagement Hours			Credits	SA		FA		Total
		L	T	P		Theory	OE/ POE	ISE	ICA	
MDP2CC4	Finite Element Method	3		2	4	60		40	25	125
MDP2CC5	Advanced Design Engineering	3			3	60		40		100
MDP2CC6	Industrial Product Design	3			3	60		40		100
MDP2CE2N*	Core Elective-II	3		2	4	60		40	25	125
MDP2CE3N*	Core Elective-III	3			3	60		40		100
MDP2SM2	Seminar-II			6	3				100	100
	Grand Total	**15**		**10**	**20**	**300**		**200**	**150**	**650**

Note: *-Students have to compulsorily undergo internship of one month after the Semester-II (in the summer vacation). The evaluation of the same will be carried out at the end of semester – III.*

Core Elective-II

Course Code	Course Title
MDP2CE21	Engineering Fracture Mechanics
MDP2CE22	Industrial Tribology
MDP2CE23	Design for Manufacturing and Assembly

Core Elective-III

Course Code	Course Title
MDP2CE31	Theory and Analysis of Composite Materials
MDP2CE32	Engineering Design Optimization
MDP2CE33	Analysis and Synthesis of Mechanisms and Machines

Note: -For Open Elective course, students should enrol for one of the minimum 8 weeks duration courses offered by SWAYAM / NPTEL platform. They should complete its assignments and appear for certificate examination conducted by SWAYAM / NPTEL. Students should pass the examination till the end of Semester IV. Based on the marks obtained in the assignments and examination; credits will be transferred in Semester IV. The list of courses will be provided by the Board of Studies.

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WALCHAND INSTITUTE OF TECHNOLOGY, SOLAPUR **(An Autonomous Institute)** **First Year M.Tech. (Design Engineering), Semester-I**			
MDP1CC1: Advanced Stress Analysis			
Teaching Scheme		**Examination Scheme**	
Lectures	3 Hours/week	**ESE**	60 Marks
Practical	----	**ISE**	40 Marks
Credits	3	**ICA**	----

Introduction:

This course is designed to equip students with a deep understanding of the mechanical behavior of solids under various loading conditions. It builds upon the fundamentals of mechanics of materials and introduces advanced analytical tools to solve complex problems in elasticity, stress analysis, and structural design. Topics such as the theory of elasticity, torsion, contact stresses, and energy methods are addressed in a rigorous and methodical manner. Students will explore real-world applications including stress analysis in pressure vessels, rotating machinery, thin-walled structures, and structural components subjected to combined loads. Analytical techniques including Airy's stress function and energy methods are emphasized for precision design and analysis.

Course Prerequisite:

Students are expected to have prior knowledge of Engineering Mechanics, Strength of Materials, Differential equations, vector calculus, linear algebra, and basic numerical methods, Basic Mechanical Design.

Course Objectives:

1. To introduce to students the Concept of three-dimensional stress and strain at a point as well stress-strain relationships for isotropic materials
2. To introduce to students the method of calculation of stresses in components of noncircular cross section subjected to unsymmetrical bending and torsional loading
3. To introduce to students the method of calculation of shear stress in thin-walled sections and determination of shear center

Course Outcomes:

After completing the course, students will be able to
1. Apply the mechanics of materials methods to engineering problems to understand structural responses to various loading conditions.
2. Formulate solutions to solid mechanics problems.
3. Comprehend current research findings as reported in journals in the field of solid mechanics.

Unit – I	**Theory of Elasticity**	**6 Hours**
Analysis of stress, analysis of strain, differential equations of equilibrium, boundary conditions, compatibility equations, Airy stress function, Bi harmonic equations		
Unit – II	**Two dimensional problems in rectangular coordinates**	**6 Hours**

Solution by polynomials, saint Venant's principle, determination of displacements, bending of a cantilever loaded at the end, bending of beam by uniform load and other cases		
Unit – III	**Pressurized Cylinders and rotating disks**	**8 Hours**
Governing equations, stresses in thick-walled cylinder under internal and external pressure, shrink fit compound cylinders, stresses in rotating flat solid disk, flat disk with central hole, disk with uniform strength		
Unit – IV	**Shear centre**	**5 Hours**
Concept of shear centre in symmetrical and unsymmetrical bending, shear centre for thin wall beam section, open section with one axis of symmetry, general open and closed section		
Unit – V	**Theory of Torsion**	**5 Hours**
Torsion of prismatic bars of solid section and thin-walled section, analogies for torsion, membrane analogy, fluid flow analogy and electrical analogy, torsion of noncircular shaft, torsion of elliptical shaft		
Unit – VI	**Contact stresses**	**5 Hours**
Geometry of contact surfaces, method of computing contact stresses and deflection of bodies in point contact, stresses for two bodies in line contact with load normal to the contact area		
Unit – VII	**Introduction to energy methods**	**5 Hours**
Energy methods for analysis of stress, strain and deflection, theorems of virtual work, Castigliano's theorem, Rayleigh Ritz method		

Reference Books

1. Theory of Elasticity - Timoshenko and Goodier, McGraw Hill Publications
2. Elasticity; Theory, Applications and Numeric's - Martin H. Sadd, Academic Press
3. Solid Mechanics: S. M. A Kaizimi - McGraw Hill Publications
4. Advanced strength of Materials - J. P. Den Harteg MGH books co Ltd.
5. Elasticity in Engineering Mechanics, Boresi A. P. and K. P. Chong, Second Edition John Wiley and Sons Publications

e-Resources

https://www.youtube.com/watch?v=_2d8YsXwm7M&list=PL35EBF66D99E7A0EC

WALCHAND INSTITUTE OF TECHNOLOGY, SOLAPUR **(An Autonomous Institute)** **First Year M.Tech. (Design Engineering), Semester-I**			

MDP1CC2: Advanced Vibrations and Acoustics

Teaching Scheme		**Examination Scheme**	
Lectures	3 Hours/week	**ESE**	60 Marks
Practical	----	**ISE**	40 Marks
Credits	3	**ICA**	----

Introduction:

Vibrations need to be considered in design of many mechanical structures. This course provides a detailed understanding of vibratory motion. It explores various types of vibrations along with the characteristics of these vibrations. It introduces various methods & devices for vibration analysis. The curriculum also deals with various methods of reduction of vibrations. The course also covers introductory part of acoustics.

Course Prerequisite:

Foundational knowledge in engineering mechanics, mathematics, and basic physics

Course Objectives:

1. To introduce students to the basics of mechanical vibrations
2. To provide students with the knowledge of determining natural frequencies of simple mechanical systems
3. To provide students with the basic knowledge of acoustics

Course Outcomes:

After completing the course, students will be able to

1. Determine natural frequencies & mode shapes of different types of vibrations of simple mechanical systems
2. Select the numerical methods to determine natural frequencies of Multi-degree of Freedom System
3. Take decision on necessity of noise control

Unit – I	**Introduction to vibrations**	**5 Hours**

Basic cause of vibrations (energy conversions), causes for vibration (earthquake & others), elimination/reduction of vibrations, definitions of terms used in vibratory motion, types of vibrations, types of damping

Unit – II	**Single Degree of Freedom system**	**5 Hours**

Natural frequency of undamped free vibrations by different methods, torsional vibrations, springs in series & parallel, damped free vibrations-viscous & coulomb damping, forced vibration with constant harmonic excitation

Unit – III	**Two Degrees of Freedom System**	**5 Hours**

Equations of motion of free vibrations, principle modes of vibration, other cases of simple tow DoFs – Two masses fixed on a tightly stretched string, double pendulum, two rotor torsional system, system with damping, un-damped forced vibrations with harmonic excitation, un-damped dynamic vibration

absorber		
Unit – IV	**Multi-degree of Freedom System**	**5 Hours**
Introduction, exact analysis: equation of motion of free undamped vibrations, representation of equations in matrix form, generalized coordinates &coordinate coupling, numerical methods: Rayleigh’s Method, Dunkerley’s method, Stodola’s method, Holzer method and matrix iteration method		
Unit – V	**Vibration of Continuous Systems**	**5 Hours**
Vibrations of string, longitudinal vibrations of bars, torsional vibrations of circular shafts, lateral vibrations of beams		
Unit – VI	**Experimental methods in Vibration analysis**	**5 Hours**
Vibration measuring instruments- displacement measuring instruments or vibrometers, velocity measuring instruments or velocity pick-ups, acceleration measuring instruments or accelerometers, Frequency measuring instruments, introduction to FFT Analyzer, vibration exciters		
Unit – VII	**Nonlinear vibrations & Random vibrations**	**5 Hours**
Introduction to nonlinear vibrations, examples of non-linear systems, phase plane –method of isoclinic, undamped free vibrations with non-linear spring forces, perturbation method, forced vibration with nonlinear spring (Duffing’s equation), random phenomena, response to random inputs		
Unit – VIII	**Introduction to Acoustics**	**5 Hours**
Basics of sound, amplitude, frequency, wave length & velocity, sound field definitions-free field, near field, far field, direct field, reverberant field, frequency analysis, one third octave band centre frequencies, quantification of sound- sound power & intensity, sound-pressure level, combining sound pressures, sources of noise & noise reduction		

Text Books

1. Mechanical Vibrations, G. K. Grover, Nem Chand & Bros, Roorkee
2. Mechanical Vibrations, V. P. Singh, Dhanpat Rai & Co. (P) Ltd., Delhi

Reference Books

1. Vibration of Mechanical Systems, C Nataraj, Cengage Learning Publications
2. Mechanical Vibrations, V. Dukkipati, J Shrinivas, PHI Learning Pvt. Ltd.
3. Theory & Practice of Mechanical Vibrations, J.S. Rao & K Gupta, New Age International Publishers
4. Mechanical Vibrations –Theory & Practice, Shrikant Bhave, Pearson India Publications
5. Mechanical Vibrations, Schaum Outlines, Tata McGraw Hill, New Delhi
6. An Introduction to Acoustics, Robert H Randall, MJP Publishers

e-Resources

https://archive.nptel.ac.in/courses/112/107/112107212/

	WALCHAND INSTITUTE OF TECHNOLOGY, SOLAPUR **(An Autonomous Institute)** **First Year M.Tech. (Design Engineering), Semester-I**	

MDP1CC3: Industrial Instrumentation

Teaching Scheme		**Examination Scheme**	
Lectures	3 Hours/week	**ESE**	60 Marks
Practical	2 Hours/week	**ISE**	40 Marks
Credits	4	**ICA**	25 Marks

Introduction:

This course provides an in-depth exploration of the instruments and systems used for measuring and controlling industrial processes. It covers the principles, design, and application of various sensors and transducers, emphasizing their role in ensuring efficiency, safety, and quality in industrial operations.

Course Prerequisite:

To effectively engage with this course, students should have a foundational understanding of Basic Electronics, Physics, Proficiency in algebra and calculus for analyzing and interpreting data.

Course Objectives:

To provide the knowledge of Pressure, Sound, Flow, Temperature, Level, Humidity, Torque, Viscosity and Vibration measurements.

Course Outcomes:

After completing the course, students will be able to
1. Classify various instruments available for measurement.
2. Select appropriate instrument for the measurement of various parameters.
3. Interpret the measurement results and cause of any possible error.

Unit – I	**Introduction to Instrumentation**	**10 Hours**

Introduction to instruments and the representation, typical applications, functional elements, classification of instruments, standards & calibration, static and dynamic characteristics of instruments

Unit – II	**Measurements by using Transducer element**	**10 Hours**

Transducer elements, intermediate elements, indicating and recording elements, displacement measurement, force measurement, torque and power measurement, pressure and vacuum measurement, flow measurement

Unit – III	**Measurement of temperature**	**5 Hours**

Temperature measurement: thermocouples, RTD, thermistors, radiation & optical, pyrometer

Unit – IV	**Measurement of vibration & Sound**	**5 Hours**
Measurement of vibration & sound, vibrometer, accelerometers, seismic instrument, sound level meter, noise analysis, signal and systems analysis, analog filters & frequency analyzers, frequency analysis, harmonic & transient testing, random force testing		
Unit – V	**Condition monitoring and Signature Analysis Applications**	**10Hours**
Condition monitoring and signature analysis applications: vibration & noise monitoring permanent monitoring system, wear behavior monitoring, corrosion monitoring, data acquisition systems, data display & storage		
ICA consists of minimum 6 assignments based on the above topics		
Reference Books		
1. Instrumentation, Measurement & Analysis, B. C. Nakra & K. K. Choudhary, Tata McGraw Hill Publications Pvt. Ltd., New Delhi 2. Instrument Devices & Systems, Rangan & Sharma, Tata McGraw Hill Publications Pvt. Ltd., New Delhi 3. Measurement Systems: Applications & Design, Earnest O Doeblin, McGraw Hill International Publications 4. Mechanical Measurement and Control, D. S. Kumar, Metropolitan Book Co. Pvt Ltd.		
e-Resources		
https://onlinecourses-archive.nptel.ac.in/noc17_ec09/preview		

WALCHAND INSTITUTE OF TECHNOLOGY, SOLAPUR
(An Autonomous Institute)
First Year M.Tech. (Design Engineering), Semester-I

CMP1RM: Research Methodology & IPR

Teaching Scheme		**Examination Scheme**	
Lectures	3 Hours/week	**ESE**	60 Marks
Practical	1 Hour/week	**ISE**	40 Marks
Credits	4	**ICA**	25 Marks

Introduction:

In the context of "Research Methodology & IPR," researchers and scholars explore how research practices intersect with intellectual property considerations. This includes understanding how to appropriately cite and use existing works, navigating copyright issues in research publication, managing intellectual property generated through research, and potentially commercializing research outcomes through patents and licensing. Overall, "Research Methodology & IPR" is a dynamic and evolving field that plays a crucial role in guiding researchers and innovators in conducting ethical, high-quality research while also respecting and protecting intellectual property rights.

Course Prerequisite:

Communication skills for writing and presenting research, analytical thinking and critical reasoning.

Course Objectives:

1. To give an overview of the research methodology and explain the technique of defining a research problem
2. To explain carrying out a literature search, its review, developing theoretical and conceptual frameworks and writing a review.
3. To explain various forms of the intellectual property, its relevance and business impact in the changing global business environment.
4. To discuss leading International Instruments concerning Intellectual Property Rights.

Course Outcomes:

After completing the course, students will be able to

1. Understand research problem formulation and approaches of investigation of solutions for research problems
2. Learn ethical practices to be followed in research and apply research methodology in case studies and acquire skills required for presentation of research outcomes
3. Discover how IPR is regarded as a source of national wealth and mark of an economic leadership in context of global market scenario
4. Summarize that it is an incentive for further research work and investment in R & D, leading to creation of new and better products and generation of economic and social benefits

Unit – I	**Introduction**	**7 Hours**
Defining research, scientific enquiry, hypothesis, scientific method, types of research, research process and steps in it. research proposals – types, contents, sponsoring agent’s requirements, ethical, training, cooperation and legal aspects		
Unit – II	**Research Design**	**7 Hours**
Meaning, need, concepts related to it, categories, literature survey and review, dimensions and issues of research design, research design process – selection of type of research, measurement and measurement techniques, selection of sample, selection of data collection procedures, selection of methods of analysis, errors in research		
Unit – III	**Research Problem**	**6 Hours**
Problem solving – types, process and approaches – logical, soft system and creative; creative problem-solving process, development of creativity, group problem solving techniques for idea generation – brain storming and Delphi method		
Unit – IV	**Nature of Intellectual Property**	**6 Hours**
Patents, designs, trade and copyright, process of patenting & development: technological research, innovation, patenting, development, international scenario international cooperation on intellectual property. procedure for grants of patents, patenting under PCT		
Unit – V	**Patent Rights**	**7 Hours**
Scope of patent rights. licensing and transfer of technology, patent information and data bases, geographical indications		
Unit – VI	**New Developments in IPR**	**6 Hours**
Administration of patent system. new developments in IPR; IPR of biological systems, computer software etc., traditional knowledge case studies, IPR		

Reference Books

1. Management Research Methodology: Integration of Principles, Methods & Techniques, Krishnaswamy, K.N., Sivakumar, Appa Iyer & Mathirajan M., - New Delhi, Pearson Education
2. Design & Analysis of Experiments, Montgomery, Douglas C. New York, John Wiley & Sons Publications
3. Research Methodology, Methods & Techniques, Kothari, C. K. New Delhi, New Age International Ltd. Publishers
4. IPR: Unleashing the Knowledge Economy, Prabuddha Ganguli, Tata McGraw Hill Publications
5. Research Design: Qualitative, Quantitative and Mixed Methods Approaches, John W Creswell, Sage Publications Pvt Ltd. 3rd Edition.
6. Research Methodology: A Step-by-Step Guide for beginners, Ranjit Kumar, 2nd Edition, SAGE Publications Ltd.
7. Resisting Intellectual Property, Halbert, Taylor & Francis Ltd.
8. Intellectual Property in New Technological Age, Robert P. Merges, Peter S. Menell, Mark A. Lemley, Clause 8 Publishing

e-Resources
https://onlinecourses.swayam2.ac.in/ntr24_ed53/preview

	WALCHAND INSTITUTE OF TECHNOLOGY, SOLAPUR **(An Autonomous Institute)** **First Year M.Tech. (Design Engineering), Semester-I**

Core Elective-I
MDP1CE11: Reliability Engineering

Teaching Scheme		**Examination Scheme**	
Lectures	3 Hours/week	**ESE**	60 Marks
Practical	2 Hours/week	**ISE**	40 Marks
Credits	4	**ICA**	25 Marks

Introduction:

Reliability is a critical aspect of any system's performance, ensuring its ability to function consistently and flawlessly over time. This subject cover's introduction to Reliability and Failure Data Analysis, analysing failure patterns to improve system resilience, Key Reliability measures, such as Mean Time To Failure (MTTF), Mean Time Between Failures (MTBF), Failure Rate, Various reliability models, evaluation of reliability systems, availability and maintainability, reliability testing are also included in the syllabus.

Course Prerequisite:

Strong foundation in engineering mathematics including calculus, linear algebra, and differential equations. Basic understanding of probability theory and statistics (e.g., distributions, hypothesis testing). Familiarity with engineering design principles and system analysis. Exposure to control systems or quality engineering is helpful but not mandatory.

Course Objectives:

1. To summarize reliability engineering and its management throughout the product life cycle.
2. To perform reliability engineering analysis.
3. To compute reliability engineering parameters and estimates for applications in mechanical devices and manufacturing environments.

Course Outcomes:

After completing the course, students will be able to

1. Apply probability theory, reliability measures, and failure models to analyze reliability characteristics of engineering components and systems.
2. Analyze failure data, reliability block diagrams, redundancy models, and maintainability parameters for evaluating reliability and availability of engineering systems.
3. Evaluate reliability design, testing, and maintenance techniques including FMECA, fault tree analysis, accelerated life testing, and repair-replacement decisions for reliable system performance.

Unit – I	**Introduction**	**4 Hours**

Brief history, concepts, terms and definitions, applications, the life cycle of a system, concept of failure, typical engineering failures and their causes, theory of probability and reliability, rules of probability, random variables, discrete and continuous probability distributions

Unit – II	Failure Data Analysis	4 Hours
Data collection and empirical methods, estimation of performance measures for ungrouped complete data, grouped complete data, analysis of censored data, fitting probability distributions graphically (exponential and Weibull) and estimation of distribution parameters		
Unit – III	**Reliability Measures**	**6 Hours**
Reliability Function–R(t), Cumulative Distribution Function (CDF)–F(t), Probability Density Function (PDF) – f(t), Hazard Rate Function-λ(t), Mean Time To failure (MTTF) and Mean Time Between Failures (MTBF), Median Time To Failure (TMED), MODE (TMODE), variance (σ2) and standard deviation (σ), typical forms of hazard rate function, bathtub curve and conditional reliability		
Unit – IV	**Basic Reliability Models**	**6 Hours**
Constant Failure Rate (CFR) model, failure modes, renewal and Poisson process, two Parameter exponential distribution, redundancy with CFR model, time-dependent failure Models, Weibull, Rayleigh, normal and lognormal distributions, burn-in screening for Weibull, redundancy, three parameter Weibull, calculation of R(t), F(t), f(t), λ(t), MTTF, TMED, TMODE, σ2 and σ for above distributions		
Unit – V	**Reliability Evaluation of Systems**	**6 Hours**
Reliability block diagram, series configuration, parallel Configuration, mixed configurations redundant systems, high level versus low level redundancy, Kout-of-n redundancy, and complex configurations, network reduction and decomposition methods, cut and tie set approach for Reliability evaluation		
Unit – VI	**Maintainability and Availability**	**4 Hours**
Concept of maintainability, measures of maintainability, Mean Time To Repair (MTTR), analysis of downtime, repair time distributions, stochastic point processes, maintenance concept and procedures, availability concepts and definitions, important availability measures		
Unit – VII	**Design for Reliability and Maintainability**	**6 Hours**
Reliability design process and design methods, reliability allocation, Failure Modes Effects and Criticality Analysis (FMECA), fault tree and success tree methods, symbols used, maintainability design process, quantifiable measures of maintainability, repair versus replacement		
Unit – VIII	**Reliability Testing**	**4 Hours**
Product testing, reliability life testing, burn-in testing, and acceptance testing, accelerated life testing and reliability growth testing		
ICA consists of minimum 8 Assignments/experiments based on the above syllabus.		

Reference Books

1. An Introduction to Reliability and Maintainability Engineering, Charles E. Ebling, 2004, Tata McGraw Hill Education Private Limited, New Delhi.
2. Reliability Engineering, L. S. Srinath, 1991, East West Press, New Delhi.
3. Reliability Engineering: Theory and Practice, Alessandro Birolini, 2010, Springer.
4. Life cycle reliability engineering, Guangbin Yang, 2007, John Wiley and Sons Publications

5. Reliability evaluation of engineering systems: Concepts and techniques, Roy Billiton and Ronald Norman Allan, Springer.
6. Practical Reliability Engineering, Patrick D. T. O'Conner, David Newton, Richard Bromley, 2002, John Wiley and Sons Publications
7. Case studies in Reliability and Maintenance, W. R. Blischke, D.N.P. Murthy, 2003, John Wiley and Sons Publications
8. Maintenance, Replacement and Reliability: Theory and Applications, Andrew Kennedy, Skilling Jardine, Albert H. C. Tsang, CRC/Taylor and Francis

e-Resources

https://onlinecourses.nptel.ac.in/e-learning/preview/noc23_ge20

WALCHAND INSTITUTE OF TECHNOLOGY, SOLAPUR
(An Autonomous Institute)
First Year M.Tech. (Design Engineering), Semester-I

Core Elective –I
MDP1CE12: Mechanical System Design

Teaching Scheme		**Examination Scheme**	
Lectures	3 Hours/week	**ESE**	60 Marks
Practical	2 Hours/week	**ISE**	40 Marks
Credits	4	**ICA**	25 Marks

Introduction:

Mechanical system design is a critical process in engineering that involves the creation of efficient, reliable, and cost-effective mechanical systems for various applications. This design process encompasses several stages, from concept development to detailed design, prototyping, testing, and implementation. Here are the key steps involved in mechanical system design such as Requirements Gathering, Conceptual Design, Preliminary Analysis, Detailed Design, Prototyping and Testing, Iterative Design Improvement, Design for Manufacturing (DFM) and Design for Assembly (DFA), Documentation and Production, Installation and Commissioning. Throughout the entire mechanical system design process, collaboration between different engineering disciplines, such as mechanical, electrical, and controls engineering, is often required, especially for more complex systems involving automation or integration with other technologies. Effective communication and coordination among team members are crucial to successful mechanical system design and implementation.

Course Prerequisite:

Fundamental knowledge of mechanics, materials science, and machine design. Strength of Materials / Mechanics of Materials, Engineering Mechanics (Statics and Dynamics), Basics of Machine Design and Theory of Machines, Manufacturing Processes and Materials Technology

Course Objectives:

1. Allow students to learn about designing mechanical systems.
2. Enable students to practically apply the fundamental knowledge gained in different applications.

Course Outcomes:

After completing the course, students will be able to
1. State different system theories and how to model it.
2. Apply different optimization methods.
3. Perform system evaluation and system simulation techniques.

Unit – I	**Engineering process and System Approach**	**4 Hours**

Basic concepts of systems, attributes characterizing a system, system types, application of system concepts in engineering, advantages of system approach, problems concerning systems, concurrent engineering

Unit – II	**Problem Formulation**	**4 Hours**
Nature of engineering problems, need statement, hierarchical nature of systems, hierarchical nature of problem environment, problem scope and constraint		
Unit – III	**System Theories**	**4 Hours**
System analysis, black box approach, state theory approach, component integration approach, decision process approach		
Unit – IV	**System modelling**	**4 Hours**
Need of modeling, model types and purpose, linear systems, mathematical modelling concept		
Unit – V	**Graph Modeling and Analysis**	**4 Hours**
Graph Modeling and analysis process, path problem, Network flow problem		
Unit – VI	**Optimization Concepts**	**4 Hours**
Optimization processes, selection of goals and objectives-criteria, methods of optimization, analytical, combinational, subjective		
Unit – VII	**System Evaluation**	**4 Hours**
Feasibility assessment, planning horizon, time value of money, financial		
Unit – VIII	**Calculus Method for Optimization**	**4 Hours**
Model with one decision variable, model with two decision variables, model with equality constraints, model with inequality		
Unit – IX	**Decision Analysis**	**4 Hours**
Elements of a decision problem, decision making, under certainty, uncertainty risk and conflict Probability, density function, expected monetary value, utility value, Baye's theorem		
Unit – X	**System Simulation**	**4 Hours**
Simulation concepts, simulation models, computer application in simulation, spread sheet Simulation, Simulation process, problem definition, input model construction and solution, limitation of simulation approach		
ICA consists of minimum 8 practical/assignments based on the above syllabus.		

Reference Books

1. Mechanical System Design- Siddiqui, Manoj Kumar Singh; New Age International Publications
2. An Introduction to Engineering Design Method- V Gupta and PN Murthy, TMH Publications, New Delhi
3. System Analysis and Project Management- Devid I Cleland, William R King, McGraw Hill Publications

e-Resources	
1.	https://nptel.ac.in/courses/110104074?utm
2.	https://nptel.ac.in/courses/108102044?utm

	WALCHAND INSTITUTE OF TECHNOLOGY, SOLAPUR **(An Autonomous Institute)** **First Year M.Tech. (Design Engineering), Semester-I**

Core Elective –I
MDP1CE13: Computer Aided Design

Teaching Scheme		**Examination Scheme**	
Lectures	3 Hours/week	**ESE**	60 Marks
Practical	2 Hours/week	**ISE**	40 Marks
Credits	4	**ICA**	25 Marks

Introduction:

This course aims to provide students with a conceptual understanding of the principles underlying CAD systems and their practical implementation. By the end of this course, students will possess a solid foundation in CAD principles, 2D and 3D transformations, geometric modeling approaches, mathematical representations, and the fundamentals of FEM. The course will let students to create, analyse, and optimize designs efficiently, making you well-equipped for real-world engineering and design challenges

Course Prerequisite:

Fundamentals of Engineering Graphics and Drawing, Knowledge of Engineering Mechanics and Materials Science, Basic familiarity with 2D and 3D modeling concepts.

Course Objectives:

1. To understand the basic analytical fundamentals that are used to create and manipulate geometric models in computer programs.
2. To visualize how the components looks like before its manufacturing or fabrication
3. To understand the different geometric modeling techniques like solid modeling, surface modeling, feature based modeling etc.

Course Outcomes:

After completing the course, students will be able to

1. Describe principles of CAD systems, and implement these principles, and its connections to CAM and CAE systems.
2. Apply 2D, 3D transformations and projection transformations to solve mechanical engineering problems
3. Create automated solid model using CAD Customization and understand CAD/CAM data exchange formats

Unit – I	**CAD Hardware and Software Systems**	**4 Hours**

CAD hardware and software, types of systems and system considerations, input and output devices, hardware integration and networking, hardware trends, software modules

Unit – II	**Computer Graphics and Geometric Transformations**	**8 Hours**

Computer graphics introduction, transformation of geometric models: translation, scaling, reflection, rotation, homogeneous representation, concatenated transformations; mappings of geometric models, translational mapping rotational mapping, general mapping, mappings as changes of coordinate

system; inverse transformations and mapping		
Unit – III	**Geometric Modeling and Curve Representation**	**8 Hours**
Projections of geometric models, orthographic projections, geometric modeling, curve representation: parametric representation of analytic curves, parametric representation of synthetic curves, curve manipulations. surface representation		
Unit – IV	**Computer Communication and Networking**	**4 Hours**
Computer communications, principle of networking, classification networks, network wring, methods, transmission media and interfaces, network operating systems		
Unit – V	**Solid Modeling Techniques and Applications**	**8 Hours**
Fundamentals of solid modeling, Boundary Representation (B-REP), Constructive Solid Geometry (CSG), sweep representation, Analytic Solid Modeling (ASM), other representations, solid manipulations, solid modeling based applications: mass properties calculations, mechanical tolerance		
Unit – VI	**Finite Element Modeling and System Simulation**	**7 Hours**
Finite element modeling and finite element analysis, finite element modeling, mesh generation mesh requirements, semiautomatic methods, fully automatic methods, design and engineering applications, system simulation, need of simulation, areas of applications, when simulation is appropriate tool/not appropriate, concept of a system, components of a system, discrete and continuous systems, model of a system, type of models, types of simulation approaches		
ICA consists of minimum 6 practical/assignments based on the above syllabus.		

Reference Books

1. CAD / CAM Theory and Practice, Ibrahbim Zeid, McGraw-Hill Publications
2. Computer Aided Engineering and Design, Jim Browne, New Age International Publications
3. CAD / CAM / CIM, P. Radhakrishnan / V. Raju / S. Subramanyam.
4. CAD / CAM principles and applications, P. N. Rao, Tata McGraw-Hill Publications
5. Mathematical Elements for Computer Graphics, Rogers / Adams, McGraw Hill Education Publications
6. Principles of Computer Aided Design, Rooney and Steadman, Aug. 1993, UCL Press in association with the Open University.
7. 7.Discrete-Event System Simulation, Jerry Banks / John Carson / Barry Nelson / David Nicol, Pearson Publications

e-Resources

1. https://nptel.ac.in/courses/112103174?utm
2. https://nptel.ac.in/courses/112103174?utm

WALCHAND INSTITUTE OF TECHNOLOGY, SOLAPUR
(An Autonomous Institute)
First Year M.Tech. (Design Engineering), Semester-I

MDP1SM1: Seminar- I

Teaching Scheme		**Examination Scheme**	
Lectures	----	**ESE**	----
Practical	4 Hours/week	**ISE**	----
Credits	2	**ICA**	50 Marks

Course Objectives:

1. To empower the student to learn beyond what is taught in class by reviewing literature available at large.
2. To review research papers periodicals, magazines and review publications on the internet and in other electronic resources.
3. To present views coherently to produce a presentation concisely with the surveyed information under the direction of the research guide.

Course Outcomes:

After completing the course, students will be able to

1. Demonstrate the ability to identify relevant information, define key concepts, and articulate topics clearly.
2. Develop persuasive speech, present information in a compelling, well-structured, and logical sequence, respond respectfully to opposing ideas, show depth of knowledge of complex subjects, and develop their ability to synthesize, evaluate and reflect on information.
3. Use appropriate methodologies, test the strength of their thesis statement, show insight into a topic, appropriate signposting, and clarity of purpose.

Topic Selection:
Topic should be based on the literature survey on any topic relevant to Design Engineering. It is desirable that the selected topic will include but not restricted to the discipline of work for the final year thesis. The scope will include Survey of patents, Research journals books and databases, Field survey and site visit reports, Communication from experts.
Report:
Each student has to prepare a write-up of about 25 to 50 pages. The report typed on A4 sized sheets and bound in the necessary format, should be submitted after approved by the guide and endorsement of the Head of Department.
Seminar Delivery:
The student has to deliver a seminar talk in front of the teachers of the department and his classmates. The Guide based on the quality of work and preparation and understanding of the candidate shall do an assessment of the seminar.
Guidelines for Seminar I report writing:
Interpretation and report writing – Techniques of interpretation – Precautions in interpretation – Significance of report writing – Different steps in report writing – Layout of research report – Mechanics of writing research report– Layout and format – Style of writing – Typing – References – Tables – Figures – Conclusion – Appendices.

	WALCHAND INSTITUTE OF TECHNOLOGY, SOLAPUR **(An Autonomous Institute)** **First Year M.Tech. (Design Engineering), Semester-I**

CMP1AC: Yoga for Stress Management

Teaching Scheme		**Examination Scheme**	
Lectures	2 Hours/week	**ESE**	----
Practical	----	**ISE**	50 Marks
Credits	----	**ICA**	----

Introduction:

This course intends to provide you with a fair understanding about the phenomenon of stress and its effective management. The course walks you through the causative aspects of stress from both eastern and western perspective, its impact on our body and mind, and ways of meeting the challenges of stress. This course presents you with information on the stress physiology, its biological pathway, stress response, and its relationship with sleep and diet. Also provide an overview of stress assessment tools and research evidence for the yogic management of stress. Further provide you stress management approaches based on the knowledge system of yoga, from theoretical and practical perspective for its effective management.

Course Prerequisite:

No special pre-requisites required

Course Objectives:

1. To train the students in the fields of Yoga and Stress management
2. To promote awareness for positive health and personality development in the student through Yoga
3. To achieve overall Good Health of Body and Mind.
4. To increase the levels of happiness.

Course Outcomes:

After completing the course, students will be able to

1. Manage stress through breathing, awareness, meditation, and healthy movement.
2. Achieve overall health of body and mind by overcoming stress
3. Use their bodies in a healthy way and perform well in sports and academics.
4. Build concentration, confidence, and positive self-image.

Unit – I	Stress & yoga philosophy	**3 Hours**
Unit – II	Managing stress	**3 Hours**
Unit – III	Worthy tips for gracious living	**3 Hours**

Unit – IV	Panch kosha, trikaya & abha (five sheaths, three bodies &aura)	**3 Hours**
Unit – V	Antahkarana chatusthya aur nigrah (four-fold mind & its control)	**3 Hours**
Unit – VI	Introduction to yoga	**3 Hours**
Unit – VII	Conditioning the mind	**3 Hours**
Unit – VIII	Pranayama (breath control)	**3 Hours**
Unit – IX	Dhaarna, dhyana & samadhi (concentration, meditation & super-consciousness)	**3 Hours**

Text Books

Acharya Yatendra Yoga & Stress Management, DivyaYog Niketan Trust, Finger-Print! Life, An imprint of Prakash Books India Pvt. Ltd., New Delhi

Reference Books

1. H R Nagendra and R Nagarathna. Yoga for Promotion of Positive Health. Swami Vivekananda Yoga Prakashana.
2. Contrada, R., & Baum, A. (Eds.). The handbook of stress science: Biology, psychology, and health. Springer Publishing Company.
3. Van den Bergh, O. Principles and Practice of Stress Management. Guilford Publications.
4. Swami Muktibodhananda, Hatha Yoga Pradipika, , Bihar School of Yoga
5. Swami Satyananda Saraswati, Four Chapters on Freedom, Bihar School of Yoga
6. Swami Tapasyananda, Srimad Bhagavat Gita, Sri Ramakrishna Math

e-Resources

https://onlinecourses.swayam2.ac.in/aic23_ge10/preview

	WALCHAND INSTITUTE OF TECHNOLOGY, SOLAPUR **(An Autonomous Institute)** **First Year M.Tech. (Design Engineering), Semester-II**

MDP2CC4: Finite Element Method

Teaching Scheme		**Examination Scheme**	
Lectures	3 Hours/week	**ESE**	60 Marks
Practical	2 Hours/week	**ISE**	40 Marks
Credits	4	**ICA**	25 Marks

Introduction:

The Finite Element Method (FEM) has emerged as an essential and irreplaceable tool within the domains of engineering and science. It enables us to address intricate problems that would otherwise be impractical or prohibitively expensive to solve using conventional analytical approaches. This course caters to a diverse audience, including aspiring engineers, experienced researchers, and individuals with a keen interest in comprehending the intricacies of computational simulations. By undertaking this course, you will acquire fundamental theoretical knowledge and invaluable practical skills required to immerse oneself in the intricacies of FEM.

Course Prerequisite:

Strong foundation in Linear Algebra and Calculus (especially partial differential equations), Basics of Engineering Mechanics (Statics and Dynamics), Understanding of Strength of Materials / Mechanics of Materials, Familiarity with numerical methods and computer programming (MATLAB, Python, or similar) is highly beneficial.

Course Objectives:

1. To learn basic principles of finite element analysis procedure.
2. To learn the theory and characteristics of finite elements that represent engineering structures.
3. To learn and apply finite element solutions to structural, thermal, dynamic problem to develop the knowledge and skills needed to effectively evaluate finite element analyses.

Course Outcomes:

After completing the course, students will be able to

1. Classify a given problem based on its dimensionality as 1-D, 2-D, or 3-D, time-dependence as Static or Dynamic, Linear or Non-linear.
2. Develop system level matrix equations from a given mathematical model of a problem.
3. Use FEA software.

Unit – I	**Introduction**	**4 Hours**

Analytical techniques in solid mechanics and fluid mechanics, numerical techniques such as FEM, BEM, FDM and FVM, computational mechanics and engineering experimentation, overview of CAE and major CAE software.

Unit – II	**Mathematics for Finite Element Methods**	**6 Hours**

Matrix algebra, vector, tensors, linear algebra, PDE, ODE, variation calculus, weighted residual method

Unit – III	Finite Element Analysis Concepts	10 Hours
a. Energy techniques in mechanics, concept of functional, Rayleigh dimensional bar element, one dimensional thermal element b. Governing differential equations, weighted residual methods strong and weak form, one dimensional bar element, one dimensional thermal element c. Types of finite element formulation, the fem process, interpretation of fem, fem history and evolution		
Unit – IV	**FEM Modeling**	**10 Hours**
a. Direct stiffness method, DOF, nodes, elements, boundary conditions, assembly and solution of displacement equations b. Shape functions, derivation of shape functions for 1d, 2d and 3d elements, polynomial, Hermite polynomial and Lagrange polynomial shape functions, convergence of shape functions c. Iso-parametric formulation: basic concept, iso-parametric elements, sub and super parametric elements, coordinate systems, mapping, assembly of equations numerical integration d. Finite elements: 1d, 2d, 3d elements, element classification, mesh refinement, mesh validity checks, sub modeling and sub structuring		
Unit – V	**Applications of FEM to Engineering problems (Software based course)**	**10 Hours**
a. Structural Analysis: static analysis, buckling analysis, modal analysis, transient analysis, spectrum analysis, nonlinear analysis (geometric non-linearity, material non-linearity and contact non-linearity) b. Thermal Analysis: conductive, convective and radiation analysis c. Coupled Field Analysis, Fatigue analysis, CFD (elementary level)		
ICA consists of minimum 8 practical/assignment based on the above syllabus		

Reference Books

1. Finite Element Procedures, Klaus-Jurgen Bathe, PHI Publication
2. The Finite Element Method: Its Basis and Fundamentals O. C. Zienkiewicz Elsevier Publication
3. A First Course in Finite Element Methods, Darryl Logan, Cengage Publication
4. An Introduction to the Finite Element Method, J. N. Reddy, McGraw Hill Publication
5. Concept and Applications on Finite Element Analysis, Cook, Malkas, Plesha, Wiley Publication
6. The Finite Element Method in Engineering, S. S. Rao, Pergamon Publication
7. A text book of Finite Element Analysis, P. Sheshu, PHI Publication
8. Introduction to Finite Elements in Engineering, Chandrupatla, PHI Publication

e-Resources

https://onlinecourses.nptel.ac.in/noc21_me31/preview

WALCHAND INSTITUTE OF TECHNOLOGY, SOLAPUR (An Autonomous Institute) First Year M.Tech. (Design Engineering), Semester-II

MDP2CC5: Advanced Design Engineering

Teaching Scheme		**Examination Scheme**	
Lectures	3 Hours/week	**ESE**	60 Marks
Practical	----	**ISE**	40 Marks
Credits	3	**ICA**	----

Introduction:

This course covers advanced concepts in mechanical design of components & systems. It explores the areas of high-speed cam design, tribology. It covers design of various types of bearings. It addresses Design for Manufacturing and Assembly (DFMA) principles. It introduces the concept of reliability in design & fatigue analysis.

Course Prerequisite:

Basic knowledge of design of mechanical components & systems

Course Objectives:

1. To introduce students to the basics of high-speed cams.
2. To inform students about the basics of tribology.
3. To provide information about various types of lubrication & bearings.
4. To make aware students about the principles of DFMA, reliability & fatigue.

Course Outcomes:

After completing the course, students will be able to
1. Select proper type of high-speed cam for a particular application with consideration of wear.
2. Compare different types of lubrication& bearings.
3. Choose proper type of hydrostatic bearing for any application & use principles of DFMA.
4. Determine reliability in the design of components or systems.
5. Design the components considering fatigue failure.

Unit – I	**Design of high-speed cams**	**5 Hours**

Types of cams, types of motion constraints – CEP, CPM, SVAJ diagrams, double dwell cam design, naïve cam design, the fundamental law of cam design, cam design –SHM, cycloidal displacement, combined functions, polynomial functions

Unit – II	**Introduction to tribology**	**5 Hours**

Definition of tribology, tribology in industry, tribology in design, friction, wear characterization, lubrication, Newton’s law of viscosity, effect of pressure & temperature on viscosity

Unit – III	**Hydrodynamic lubrication**	**5 Hours**

Principle of hydrodynamic lubrication, mechanism of pressure development, lubrication regimes, Reynold's equation with assumptions		
Unit – IV	**Hydrodynamic journal bearing**	**5 Hours**
Nomenclature, principle of hydrodynamic lubrication for journal bearing, infinitely long journal bearing- pressure distribution, infinitely short (narrow) journal bearing – pressure distribution, load carrying capacity, volume flow rate in an axial direction		
Unit – V	**Introduction to hydrostatic bearings**	**4 Hours**
Advantages, limitations & applications of hydrostatic bearings, viscous flow through rectangular slot, hydrostatic step bearing, bearing materials, seals & gaskets		
Unit – VI	**Design for Manufacturing and Assembly**	**7 Hours**
History of DFMA, Design for Manufacturing (DFM), Design for Assembly (DFA), DFMA process, Softwares for DFMA		
Unit – VII	**Introduction to Reliability in Design**	**6 Hours**
Definitions of reliability, failure-data analysis- failure data, mean failure rate, mean time to failure, mean time between failure, system reliability – series configuration, parallel configuration, mixed configuration, numerical, Hazard models – constant hazard, linearly increasing hazard, the Weibull model, reliability improvement		
Unit – VIII	**Fatigue Analysis**	**7 Hours**
Introduction, fatigue strength, factors affecting fatigue behaviour, high cycle and low cycle fatigue, cumulative damage in fatigue and fatigue under complex Stresses		

Text Books

1. Theory of Machines and mechanisms by J. E. Shigley (TMGH), Oxford University Press.
2. Industrial Tribology, R. B.Patil, Tech-Max Publications
3. Concepts in Reliability engineering by L. S. Srinath, East West press Pvt. Ltd.

Reference Books

1. Mechanical Engineering Design by J. E. Shigley TMGH, Publications
2. Kinematics & Dynamics of Machinery, R. C. Norton, Tata McGraw Hill Publications
3. Introduction to Tribology of bearing by Majumdar, Wheeler Publication
4. Theory of Hydrodynamic Lubrication by Pinkus „O' Stemitch
5. Tribology in Industry by Susheel Kumar Srivastav, S. Chand& Co.
6. Reliability Engineering by E. Balguruswamy, TMGH Publications
7. Mechanisms and Design of cam mechanisms by Fan Y Chen, Pergamon Press inc.

e-Resources

1. https://nptel.ac.in/courses/112101096?utm
2. https://nptel.ac.in/courses/112107142?utm

	WALCHAND INSTITUTE OF TECHNOLOGY, SOLAPUR **(An Autonomous Institute)** **First Year M.Tech. (Design Engineering), Semester-II**

MDP2CC6: Industrial Product Design

Teaching Scheme		**Examination Scheme**	
Lectures	3 Hours/week	**ESE**	60 Marks
Practical	----	**ISE**	40 Marks
Credits	3	**ICA**	----

Introduction:

It has been observed that the most successful economies are based on innovation and creativity led entrepreneurship by successful product and service to consumer and industry. The main objective of the course is to acquaint the students with the practical knowledge regarding conceptualization, design and development of industrial and consumer product using modern tools. The need and importance of industrial design, new product, the product life cycle, the product design process, the application of economical consideration in product design, various product design tools such as CAD, QFD, concurrent approach should able to apply for specific examples/ case studies in included in the syllabus. The concept of Ergonomics and Aesthetics in context of the industrial product design should be understood with the help of case studies.

Course Prerequisite:

Engineering Graphics and Design Fundamentals, Engineering Mechanics and Materials, Basics of CAD and Design Software, Manufacturing Processes, Basic Aesthetic and Ergonomic Principles (for user-centric product design).

Course Objectives:

1. To impart knowledge and skill set required for industrial design profession which includes ergonomic and aesthetic in design.
2. To enable student to understand industrial design practices includes new product design.
3. To enable student to understand role of design engineer which impact at national and international level.

Course Outcomes:

After completing the course, students will be able to
1. Apply product design process to industrial and consumer products.
2. Use aesthetic and ergonomics concept in the product design.
3. Implement critical and creative thinking in the design of products.

Unit – I	**Introduction**	**6 Hours**

Need of industrial design, concept development process, design and development process of industrial products, assessing the quality of industrial design, problems faced by industrial designer, types of models used in industrial design-clay studies, mock ups, scale models, prototypes

Unit – II	**Industrial Product Design**	**6 Hours**

Design of industrial and consumer products, setting specification, requirements and rating, their importance in the design, study of market requirements and manufacturing aspects of industrial designs, challenges of product development		
Unit – III	**Aesthetic and Ergonomic Concepts**	**4 Hours**
Concept of unity and order with variety, concept of purpose, style, and environment, aesthetic expressions of symmetry, balance, contrast continuity, proportion		
Unit – IV	**Aesthetic and Ergonomic Concepts**	**4 Hours**
Mechanics of seeing, psychology of seeing. Influence of line and form, man-machine relationship, use and limitations of anthropometric data, aspects of ergonomic design of machine tools, testing equipments, instruments, automobiles, process equipments		
Unit – V	**New Product Development**	**4 Hours**
Initiation, Idea collection, creative design; brain storming; creative thinking; creative development, inventiveness; conception design, function and use, legal standard requirement, international standards, prototype design pre-production, inspection		
Unit – VI	**Economic Considerations**	**6 Hours**
Selection of material, Design for Production (DFP), impact of DFP on other factors, use of standardization, value analysis and cost reduction, break even analysis		
Unit – VII	**Design for Environment**	**4 Hours**
Need, guidelines, product life cycle assessment, techniques to reduce environmental impact		
Unit – VIII	**Modern approaches to product design**	**6 Hours**
Concurrent design, Quality Function Deployment (QFD), computer aided industrial design, rapid prototyping		
ICA consists of minimum 6 practical/assignment based on the above syllabus		

Reference Books

1. Product Design and development – Karl T. Ulrich, Steven D. Eppinger and Anita Goyal, McGill Education Publications
2. Product Design – Kevin Otto and Kristin Wood, Pearson Education Publications
3. Product Design and Manufacture- A. K. Chitale and R. C. Gupta, PHI Learning, Publications
4. Industrial Design for Engineers – W. H. Mayall, London Life books Ltd.

e-Resources

https://onlinecourses.nptel.ac.in/e-learning/preview/noc21_me66

	WALCHAND INSTITUTE OF TECHNOLOGY, SOLAPUR **(An Autonomous Institute)** **First Year M.Tech. (Design Engineering), Semester-II**

Core Elective II
MDP2CE21: Engineering Fracture Mechanics

Teaching Scheme		**Examination Scheme**	
Lectures	3 Hours/week	**ESE**	60 Marks
Practical	2 Hours/week	**ISE**	40 Marks
Credits	4	**ICA**	25 Marks

Introduction:

Fracture mechanics is the mechanical analysis of materials containing one or more cracks to predict the conditions when failure is likely to occur. It is the field of mechanics concerned with the study of the propagation of cracks in materials. It uses methods of analytical solid mechanics to calculate the driving force on a crack and those of experimental solid mechanics to characterize the material's resistance to fracture.

Course Prerequisite:

Mechanics of Materials / Strength of Material, Engineering Mathematics, Knowledge of stress analysis, elasticity, and failure theories, Understanding of material structure, microstructure, and failure modes.

Course Objectives:

1. Study the concept of failure in members with pre-existing flaws.
2. Acquire basic skills, to work professionally as an engineer for applying fracture mechanics theory and to calculate stress areas and the "energy release rate" around crack tips and crack growth due to fatigue.
3. Examine Failure of structural components from the mechanics and micro structural points of view.
4. Learn to employ modern numerical methods to determine critical crack sizes and fatigue crack propagation rates in engineering structures.

Course Outcomes:

After completing the course, students will be able to

1. Describe structural failure modes and the principles of fracture mechanics.
2. Calculate crack propagation rates and stress intensity factors for various crack geometries.
3. Analyse crack tip plasticity and evaluate the influence of plate thickness on fracture behaviour.
4. Assess fatigue behaviour and solve problems related to high-temperature material performance.

Unit – I	**Introduction**	**4 Hours**
Kinds of failure, historical aspects, brittle and ductile fracture, modes of fracture failure		
Unit – II	**Fracture Criteria**	**8 Hours**
Griffith criterion, Irwin's fracture criterion, stress intensity approach, stress intensity factor, surface energy, energy release rate, crack resistance, r curve for brittle crack, stable and unstable crack growth, critical energy release rate		

Unit – III	**Stress intensity factor**	**8 Hours**
Stress and displacement fields, edge cracks, embedded cracks, SIF for different geometry, critical stress intensity factor		
Unit – IV	**Crack tip plasticity**	**8 Hours**
Shape and size of plastic zone, effective crack length, effect of plate thickness, J-Integral. Crack tip opening displacement, test methods for determining critical energy release rate, critical stress intensity factor, J- Integral: clip gauge, load displacement test		
Unit – V	**Fatigue mechanics**	**8 Hours**
S-N diagram, fatigue limit, fatigue crack growth rate, Paris law, crack propagation, effect of an overload, crack closure, variable amplitude fatigue load, environment- assisted cracking, dynamic mode crack initiation and growth, various crack detection techniques		
Unit – VI	**Creep**	**4 Hours**
Creep and stress rupture, high temperature materials problems, time dependent mechanical behaviour, the creep curve, the stress rupture test, structural changes during creep		

Text Books

1. Prashant Kumar, Elements of fracture mechanics, McGraw Hill Education (I) Pvt., Ltd.
2. Broek David, Elementary Engineering Fracture Mechanics, 3rd Rev. Ed. Springer

Reference Books

1. Anderson T.L., Fracture Mechanics, 2nd Edition, CRC Press

e-Resources

https://nptel.ac.in/courses/112106065

WALCHAND INSTITUTE OF TECHNOLOGY, SOLAPUR (An Autonomous Institute) First Year M.Tech. (Design Engineering), Semester-II			
Core Elective II **MDP2CE22: Industrial Tribology**			
Teaching Scheme		**Examination Scheme**	
Lectures	3 Hours/week	**ESE**	60 Marks
Practical	2 Hours/week	**ISE**	40 Marks
Credits	4	**ICA**	25 Marks

Introduction:

This course provides an in-depth exploration of the fundamental principles and applications in the field of tribology, as well as the essential concepts related to lubrication and lubricants. By the end of this course, Student will have a comprehensive understanding of the principles and applications of tribology and lubrication, equipping with valuable knowledge for tackling real world engineering challenges and optimizing machinery performance.

Course Prerequisite:

Engineering Mechanics, Mechanics of Materials / Strength of Materials, Basics of fluid flow are essential for studying lubrication theory.

Course Objectives:

1. Understand the basic concepts, theories, and principles of tribology
2. Gain the knowledge of friction and wear
3. Understand the principle of hydrostatic lubrication, hydrodynamic lubrication and air/gas lubricated bearings
4. Understand the signification of lubrication and lubricants

Course Outcomes:

After completing the course, students will be able to

1. Explain the fundamentals of tribology, including viscosity, surface properties, and contact analysis for smooth and rough surfaces.
2. Analyse friction and wear mechanisms in metals and non-metals, along with relevant laws and measurement techniques.
3. Apply principles of hydrostatic and hydrodynamic lubrication in the design and operation of thrust and journal bearings.
4. Evaluate air/gas-lubricated bearings and the role of lubricants, considering their performance, advantages, and limitations.

Unit – I	**Introduction**	**6 Hours**

Tribology in design, tribology in industry viscosity, flow of fluids, viscosity and its variation absolute and kinematic viscosity, temperature variation, viscosity index determination of viscosity, different viscometers, tribological considerations nature of surfaces and their contact, physic mechanical properties of surface layer, geometrical properties of surfaces, methods of studying surfaces, study of contact of smoothly and rough surfaces

Unit – II	Friction and wear	6 Hours
Role of friction and laws of static friction, causes of friction, theories of friction, laws of rolling friction, friction of metals and non-metals, friction measurements, definition of wear, mechanism of wear, types and measurement of wear, friction affecting wear, theories of wear, wear of metals and non-metals		
Unit – III	**Hydrostatic lubrication**	**8 Hours**
Principle of hydrostatic lubrication, general requirements of bearing materials, types of bearing materials, hydrostatic step bearing, application to pivoted pad thrust bearing and other applications, hydrostatic lifts, hydrostatic squeeze films and its application to journal bearing, optimum design of hydrostatic step bearing		
Unit – IV	**Hydrodynamic theory of lubrication**	**8 Hours**
Principle of hydrodynamic lubrication, various theories of lubrication, Petroff's equation, Reynold's equation in two dimensions -effects of side leakage – Reynold's equation in three dimensions, friction in sliding bearing, hydro dynamic theory applied to journal bearing, minimum oil film thickness, oil whip and whirl, anti –friction bearing, hydrodynamic thrust bearing		
Unit – V	**Air/gas lubricated bearing**	**7 Hours**
Advantages and disadvantages application to hydrodynamic journal bearings, hydrodynamic thrust bearings, hydrostatic thrust bearings, hydrostatic bearing analysis including compressibility effect		
Unit – VI	**Lubrication and lubricants**	**5 Hours**
Introduction, dry friction, boundary lubrication, classic hydrodynamics, hydrostatic and elasto-hydrodynamic lubrication, functions of lubricants, types of lubricants and their industrial uses, sae classification, recycling, disposal of oils, properties of liquid and grease lubricants, lubricant additives, general properties and selection		

Text Books

1. Bearing Design in Machinery – Avrahan Harnoy, CRC Press Inc
2. Basic Lubrication Theory- A Camaron,John Wiley & Sons, Publications
3. Principles of Lubrication– A Camaron, Longman's Green Co. Ltd.
4. Theory and Practice for Engineers– D. D. Fuller, John Wiley and Sons Publications
5. Fundamentals of Tribology, Basu, SenGupta and Ahuja/PHI Publications
6. Tribology in Industry: Sushil Kumar Srivatsava, S. Chand Publications
7. Tribology – B.C. Majumdar, McGraw Hill Publications

Reference Books

1. Gas Lubrication Bearings – Grassam and Powell, Butterworths Publications
2. Theory Hydrodynamic Lubrication, Pinkush and Sterrolicht, Mc Graw Hill Publications
3. Tribology in Machine Design– T. A. Stolarski, Butterworth-Heinemann Ltd
4. Fundamental of Friction and Wear of Metals– ASM
5. Standard Hand Book of Lubrication Engg., O'Conner and Royle, McGraw Hills Publications
6. Introduction to Tribology, Halling , Wykeham Publications Ltd.
7. Lubrication, Raymond Gunther, Chilton Book Co.
8. Bearing Systems: Principles and Practice, P T Barwll, Oxford University Press

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https://nptel.ac.in/courses/112102015

WALCHAND INSTITUTE OF TECHNOLOGY, SOLAPUR
(An Autonomous Institute)
First Year M.Tech. (Design Engineering), Semester-II

Core Elective II
MDP2CE23: Design for Manufacturing and Assembly

Teaching Scheme		Examination Scheme	
Lectures	3 Hours/week	**ESE**	60 Marks
Practical	2 Hours/week	**ISE**	40 Marks
Credits	4	**ICA**	25 Marks

Introduction:

This course aims to equip students with a comprehensive understanding of design principles for manufacturability, encompassing strength, mechanical factors, mechanisms selection, evaluation methods, and process capability, including feature and geometric tolerances, assembly limits, datum features, and tolerance stacks. Additionally, it explores the crucial factors influencing form design, such as working principles, materials, manufacturing considerations, and design solutions for welded members, forgings, and castings. Furthermore, the course covers component design with an emphasis on machining and casting considerations, along with techniques for reducing environmental impact and designing for the environment, enabling students to develop sustainable and efficient engineering solutions.

Course Prerequisite:

Understanding the basics of machine elements and design principles, Knowledge of machining, casting, forming, and joining processes., Awareness of how materials influence manufacturability and assembly, Familiarity with product lifecycle and design stages.

Course Objectives:

1. To introduce the concept "Design for X" (DFX), that is well established within product development.
2. To introduce and aware students about the basic design process which based on different aspects of manufacturing as well assembly.
3. To give an idea about different criteria made on design such as machining and casting.

Course Outcomes:

After completing the course, students will be able to

1. Identify manufacturing considerations critical to the mechanical design process.
2. Apply machining principles to enhance product design for efficiency, economy, assembly, and accessibility.
3. Design cast components by optimizing parting lines, minimizing cores, and integrating Design for Manufacturing and Assembly (DFMA) principles.
4. Evaluate environmental objectives and apply Design for the Environment (DFE) methods, including life cycle assessment, for sustainable product design.

Unit – I	**Introduction**	**4 Hours**
General design principles for manufacturability: strength and mechanical factors, mechanisms selection, evaluation method, process capability: feature tolerances, geometric tolerances, assembly limits, datum features, and tolerance stacks		
Unit – II	**Factors Influencing form Design**	**8 Hours**
Working principle, material, manufacture, design - possible solutions, materials choice, influence of materials on form design, form design of welded members, forgings and castings		
Unit – III	**Component Design-I**	**8 Hours**
Machining consideration: design features to facilitate machining drills, milling cutters, keyways, doweling procedures, counter sunk screws, reduction of machined area, simplification by separation, simplification by amalgamation, design for machinability, design for economy, design for clamp ability, design for accessibility, design for assembly		
Unit – IV	**Component Design-II**	**8 Hours**
Casting consideration: redesign of castings based on parting line considerations, minimizing core requirements, machined holes, redesign of cast members to obviate cores, identification of uneconomical design, modifying the design, group technology, computer applications for DFMA.		
Unit – V	**Design for the Environment**	**6 Hours**
Introduction, environmental objectives, global issues regional and local issues, basic DFE methods, design guide lines, example application, lifecycle assessment, basic method, environmentally responsible product assessment, weighted sum assessment method, lifecycle assessment method		
Unit – VI	**Techniques to reduce environmental impact**	**6 Hours**
Techniques to reduce environmental impact, design to minimize material usage, design for disassembly, design for recyclability, design for remanufacture, design for energy efficiency, design to regulations and standards		
ICA consists of minimum 6 Assignments/experiments based on the above syllabus.		

Reference Books

1. Product Design- Kevien Otto and Kristin Wood, Pearson Publication
2. Product design and development, K.T. Ulrich and S.D. Eppinger, Tata McGraw Hill Publication
3. Design for Assembly Automation and Product Design, Boothroyd, Marcel Dekker Publication
4. Design for Manufacture handbook, Bralla, McGraw Hill Publication

e-Resources

https://nptel.ac.in/courses/107103012

WALCHAND INSTITUTE OF TECHNOLOGY, SOLAPUR
(An Autonomous Institute)
First Year M.Tech. (Design Engineering), Semester-II

Core Elective-III
MDP2CE31: Theory and Analysis of Composite Materials

Teaching Scheme		**Examination Scheme**	
Lectures	3 Hours/week	**ESE**	60 Marks
Practical	----	**ISE**	40 Marks
Credits	3	**ICA**	----

Introduction:

Theory and Analysis of Composite Materials is a specialized field of study that focuses on the properties, behaviour, design, and applications of composite materials. Composite materials are engineered materials made by combining two or more distinct materials to create a new material with improved and tailored properties. These materials offer a wide range of advantages over traditional materials due to their unique combination of characteristics.

Course Prerequisite:

Understanding stress-strain relations, axial/bending/torsional loads, Matrix algebra, differential equations, calculus, and numerical methods, Stress tensors, strain-displacement relations, elasticity, Microstructure, anisotropy, failure mechanisms.

Course Objectives:

1. Define and explain the fundamental concepts of composite materials, including matrix, reinforcement, and microstructure.
2. Study stress, strain, and deformation distribution within composite structures.
3. Learn mathematical and analytical methods to predict composite behavior under different loading conditions.
4. Identify and assess potential failure mechanisms and modes in composite materials.
5. Explain various manufacturing techniques used to create composite materials.
6. Design composite components for specific applications, considering factors such as loadbearing capacity and weight reduction.

Course Outcomes:

After completing the course, students will be able to

1. Describe the relationship between microstructure and material properties in composite materials.
2. Apply analytical methods to predict the mechanical behaviour of composites under various loading conditions.
3. Analyse stress distribution and failure mechanisms such as delamination, fibre breakage, and matrix cracking.
4. Design composite structures for specific applications, incorporating suitable manufacturing techniques and performance requirements.

Unit – I	**Introduction to Composite Materials**	**6 Hours**
Definition, classification, types of matrix material sand reinforcements, characteristics & selection, fiber composites, laminated composites, metal matrix composite, particulate composites and pre-pegs, application of composite materials		
Unit – II	**Macro-mechanical behaviour of a Lamina**	**8 Hours**
Hooke's law for different types of materials, number of elastic constants, two - dimensional relationship of compliance and stiffness matrix, stress –strain relations for plane stress in an orthotropic material, strengths of an orthotropic lamina, numerical problems		
Unit – III	**Micro-mechanical behaviour of a Lamina**	**6 Hours**
Introduction, mechanics of material approach to stiffness, elasticity approach to stiffness, evaluation of the four elastic moduli, rule of mixture, numerical problems, biaxial strength theories: maximum stress theory, maximum strain theory, numerical problems		
Unit – IV	**Macro-mechanical behaviour of Laminate**	**8 Hours**
Introduction, code, Kirch off hypothesis, classical lamination theory, a, b, and d matrices (detailed derivation) engineering constants, special cases of laminates, strength of laminate, inter-laminar stresses		
Unit – V	**Manufacturing of Ceramic materials**	**6 Hours**
Open and closed mould processing, hand lay-up techniques, bag molding and filament winding, pultrusion, pull forming, thermoforming, injection molding, types of defects		
Unit – VI	**Behaviour of Composite Materials**	**6 Hours**
Introduction, governing equations for bending and buckling of laminated plates, principles of fracture mechanics and effect of discontinuity in laminates, applications		

Text Books

1. Mechanics of Composite Materials, R.M. Jones, Taylor & Francis.
2. Mechanics of composite materials, Autar K. Kaw, CRC Press New York.
3. Composite Materials handbook, Mein Schwartz, McGraw Hill Book Company

Reference Books

1. Stress analysis of fiber Reinforced Composite Materials, Michael W, Hyer, McGraw Hill Publications
2. Composite Material Science and Engineering, Krishan K. Chawla, Springer.
3. Fiber Reinforced Composites, P.C. Mallik, Marcel Decker Publications

e-Resources

http://digimat.in/nptel/courses/video/112104229/L01.html

Core Elective-III
MDP2CE32: Engineering Design Optimisation

Teaching Scheme		**Examination Scheme**	
Lectures	3 Hours/week	**ESE**	60 Marks
Practical	----	**ISE**	40 Marks
Credits	3	**ICA**	----

Introduction:

Optimization is the process of obtaining the best result under given circumstances. In design, construction and maintenance of any engineering system, engineers have to take many technological and managerial decisions at several stages. The ultimate goal of all such decisions is either to minimize the effort required or to maximize the desired benefit. A number of optimization methods have been developed for solving different types of optimization problems. This course covers optimization problem formulation, Linear & Non-Linear Programming with number of applications in mechanical engineering. Multi objective optimization is briefly introduced.

Course Prerequisite:

Familiarity with basics of probability, statistics, and linear algebra is essential.

Course Objectives:

1. To introduce students with basic concept of optimization
2. To explore various classical optimization techniques of optimization
3. To give information about linear & nonlinear optimization problems
4. To provide knowledge of multi-objective optimization problems of real world

Course Outcomes:

At the end of the course students will be able to
1. Formulate an optimization problem
2. Solve problems related to Classical Optimization method
3. Apply techniques of optimization for linear& nonlinear programming problems
4. Differentiate between single variable & multi-variable optimization problems
5. Optimize the design of components a mechanical system

Unit – I	**Introduction to Optimization**	**4 Hours**

Need for optimization and historical development, engineering application of optimization, classification of optimization problems, formulation and statement of optimization problems

Unit – II	**Classical Optimization Methods**	**5 Hours**

Introduction, review of single and multivariable optimization techniques with or without constraints

Unit – III	**Linear Programming**	**5 Hours**
Standard form of linear programming, geometry of linear programming, solutions of system of linear simultaneous equations		
Unit – IV	**Non-Linear Programming**	**6 Hours**
One dimensional minimization method, elimination methods, unrestricted search, exhaustive search, golden section method		
Unit – V	**Non-Linear Programming (Unconstrained Optimization)**	**5 Hours**
Direct search method, random search method, grid search method, indirect search method		
Unit – VI	**Non-Linear Programming (Constrained Optimization)**	**5 Hours**
Direct methods, random search method, sequential linear programming, sequential quadratic programming		
Unit – VII	**Optimization Design of Mechanical Systems**	**5 Hours**
Case studies of optimization design of mechanical systems		
Unit – VIII	**Multi-objective Optimum Design**	**5 Hours**
Concepts and methods, genetic algorithms, weighted sum method, weighted minimum maximum method, Global optimization concepts and methods for optimum design		
Text Books		
1. Engineering Optimization – S. S. Rao, John Wiley & Sons Publications		
Reference Books		
1. Introduction to Optimization, JasbirAroa, Elsevier Academic Press 2. Optimization Theory and Applications, S S Rao, Wiley Eastern LtdPublications 3. Optimization for Engineering Design, Kalyanmoy Deb, PHI Publications 4. Optimization Concepts & Application in Engineering, Belgundu & Chandrupatla, Cambridge University Press 5. Applied Optimal Design, E J Jaug, J S Arora, McGraw HillPublications 6. Principles of Optimization Design, Papalambros & Wilde, Cambridge University Press 7. Operations Research, D. S. Hira & Gupta, S. Chand Publications		
e-Resources		
https://nptel.ac.in/courses/111105039		

	WALCHAND INSTITUTE OF TECHNOLOGY, SOLAPUR **(An Autonomous Institute)** **First Year M.Tech. (Design Engineering), Semester-II**

Core Elective-III
MDP2CE33: Analysis and Synthesis of Mechanisms and Machines

Teaching Scheme		**Examination Scheme**	
Lectures	3 Hours/week	**ESE**	60 Marks
Practical	----	**ISE**	40 Marks
Credits	3	**ICA**	----

Introduction:

Mechanisms and machines play a crucial role in various engineering applications, ranging from simple everyday devices to complex industrial systems. The synthesis and analysis of mechanisms and machines involve designing, optimizing, and understanding their performance and behavior. It includes synthesis, kinematic and dynamic analysis of machines and mechanisms. Overall, the synthesis and analysis of mechanisms and machines are essential processes in engineering to ensure the creation of efficient, reliable, and safe systems for various applications. These concepts are studied in-depth in mechanical engineering and related disciplines. Engineers use computer-aided design (CAD) software and simulation tools to model, analyze, and optimize mechanisms and machines before they are manufactured and put into operation.

Course Prerequisite:

Kinematic chains, degrees of freedom, four-bar mechanisms, cam-follower systems Mechanics of Materials, Matrix algebra, vector calculus, and differential equation, Machine Design

Course Objectives:

1. Study the basic concepts of machines and mechanisms.
2. Learn about the kinematic analysis of complex mechanisms.
3. Get acquainted with the dynamic analysis of mechanisms.
4. Study the various graphical methods for synthesis of mechanisms.
5. Learn about the various analytical methods for synthesis of planer mechanisms.
6. Study the kinematic analysis of spatial mechanisms.

Course Outcomes:

After completing the course, students will be able to

1. Describe the construction and design parameters of planar and spatial mechanisms.
2. Analyse the kinematic and dynamic behaviour of simple and complex mechanisms.
3. Evaluate mechanism performance based on motion, path, and body guidance criteria.
4. Design and synthesize mechanisms for real-life applications using analytical and matrix-based methods.

Unit – I	**Basic Concepts**	**4 Hours**

Basic Concepts: definition s and assumptions, planar and spatial mechanisms, kinematic pairs, degree of freedom

Unit – II	**Kinematic Analysis of Complex Mechanisms**	**5 Hours**
Velocity-acceleration analysis of complex mechanisms by the normal acceleration and auxiliary point methods		
Unit – III	**Dynamic Analysis of Planar Mechanisms and Curvature theory**	**11 Hours**
Inertia forces in linkages, kinetic, static analysis of mechanisms by matrix method, analysis of elastic mechanisms, beam element, displacement fields for beam element, element mass and stiffness matrices, system matrices, elastic linkage model, equations of motion, fixed and moving centrodes, inflection circle, Euler- Savary equation, Bobillier constructions, cubic of stationary curvature, Ball's point, applications in dwell mechanisms		
Unit – IV	**Graphical Synthesis of Planar Mechanisms**	**7 Hours**
Type, number and dimensional synthesis, function generation, path generation and rigid body guidance problems, accuracy (precision) points, Chebychev spacing, types of errors, graphical synthesis for function generation and rigid body guidance with two, three and four accuracy points using pole method, center point and circle point curves, Bermester points, synthesis for five accuracy points, branch and order defects, synthesis for path generation		
Unit – V	**Analytical synthesis of Planar Mechanisms**	**7 Hours**
Analytical synthesis of four-bar and slider- crank mechanism, Freudenstein's equation, synthesis for four accuracy points, compatibility condition, synthesis of four-bar for prescribed angular velocities and accelerations using complex numbers, complex numbers method of synthesis, the dyad, center point and circle point circles, ground pivot specifications, three accuracy point synthesis using dyad method, Robert Chebychev theorem, cognates		
Unit – VI	**Kinematic Analysis of Spatial Mechanisms**	**6 Hours**
Denavit- Hartenberg parameters, matrix method of analysis of spatial mechanisms		

Text Books

1. Theory of Machines and Mechanisms, A. Ghosh and A. K. Mallik, Affiliated East West Press.
2. Kinematic Synthesis of Linkages, R. S. Hartenberg and J. Denavit, McGraw -Hill Publications

Reference Books

1. Mechanism Design - Analysis and Synthesis (Vol.1 and 2), A. G. Erdman and G. N. Sandor, Prentice Hall Publications
2. Theory of Machines and Mechanisms, J. E. Shigley and J. J. Uicker, 2nd Ed., McGraw-Hill Publications
3. Design of Machinery: An I introduction to the Synthesis and Analysis of Mechanisms and Machines, Robert L. Norton, Tata McGraw-Hill Publications
4. Kinematics and Linkage Design, A. S. Hall, Prentice Hall Publications

e-Resources

http://acl.digimat.in/nptel/courses/video/112106270/L31.html

WALCHAND INSTITUTE OF TECHNOLOGY, SOLAPUR
(An Autonomous Institute)
First Year M.Tech. (Design Engineering), Semester-II

MDP2SM2: Seminar- II

Teaching Scheme		**Examination Scheme**	
Lectures	----	**ESE**	----
Practical	6 Hours/week	**ISE**	----
Credits	3	**ICA**	100 Marks

Course Objectives:

1. To empower the student to learn beyond what is taught in class by reviewing literature available at large.
2. To review research papers periodicals, magazines and review publications on the internet and in other electronic resources.
3. To present views coherently to produce a presentation concisely with the surveyed information under the direction of the research guide.

Course Outcomes:

After completing the course, students will be able to

1. Demonstrate the ability to identify relevant information, define key concepts, and articulate topics clearly.
2. Develop persuasive speech, present information in a compelling, well-structured, and logical sequence, respond respectfully to opposing ideas, show depth of knowledge of complex subjects, and develop their ability to synthesize, evaluate and reflect on information.
3. Use appropriate methodologies, test the strength of their thesis statement, show insight into a topic, appropriate signposting, and clarity of purpose.

Topic Selection: Topic should be based on the literature survey on any topic relevant to Design Engineering. At least five journal papers should be referred for topic selection. It is desirable that the selected topic may be leading to selection of a suitable topic of dissertation.

Report: Each student has to prepare a write-up of about 25 to 50 pages. The report typed on A4 sized sheets and bound in the necessary format, should be submitted after approved by the guide and endorsement of the Head of Department. It is expected that a review paper based on literature review is to be presented at least in conference.

Seminar Delivery: The student has to deliver a seminar talk in front of the teachers of the department and his classmates. It is expected to invite external examiner for the seminar. Based on the quality of content, understanding of the candidate, and seminar delivery the Guide or external examiner shall do an assessment of the seminar.

Guidelines for Seminar II report writing:

Interpretation and report writing – Techniques of interpretation – Precautions in interpretation – Significance of report writing – Different steps in report writing – Layout of research report – Mechanics of writing research report – Typing – References – Tables – Figures – Conclusion – Appendices.